Dual Radio Map-Aware Flight Strategy Optimization for UAV-Based Inspection System

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***Abstract*—This paper investigates a dual radio map-aware inspection system that employs a cellular-connected unmanned aerial vehicle (UAV) in rural environments. The UAV is tasked with executing a series of mission-critical inspection operations while transmitting the collected data in real time under the guidance of pre-constructed radio maps. To enhance inspection efficiency and ensure low-latency data feedback, a joint optimization problem is formulated to simultaneously determine the inspection sequence and flight trajectory, with the objective of minimizing both mission completion time and data transmission delay. To solve this problem, we propose a novel framework that integrates an improved genetic algorithm (IGA) with deep reinforcement learning (DRL). The IGA first derives a near-optimal inspection sequence by jointly considering flight distance, uplink communication throughput, and downlink outage duration based on ground-to-air and air-to-ground radio maps. Subsequently, the DRL module refines the continuous flight trajectory. The simulation results demonstrate that the proposed DRL-IGA outperforms benchmark schemes, achieving up to a 20% reduction in both total task completion time and transmission delay.**

***Index Terms*—Cellular-connected UAV, deep reinforcement learning, dual radio map, trajectory optimization.**

I. Introduction

Unmanned aerial vehicles (UAVs) have attracted considerable research and industrial interest in recent years due to their high maneuverability, low cost, and ease of deployment, enabling a wide range of applications such as cargo transportation, emergency communications, and aerial surveillance [1]–[3]. Among these, cellular-connected UAVs, functioning as aerial users within terrestrial cellular networks, represent a promising paradigm. By leveraging existing ground base stations (GBSs), they overcome the range limitations of direct line-of-sight (LoS) communications, thereby ensuring reliable command and control links and supporting high-throughput data transmission [4]. However, integrating UAVs into cellular networks presents substantial challenges. Most existing GBSs are optimized for terrestrial users through antenna downtilts, resulting in poor signal reception and frequent handovers for high-altitude UAVs, thereby undermining communication reliability. Furthermore, downlink interference from non-associated GBSs degrades communication quality and leads to coverage holes [5], thereby constraining the overall performance and reliability of cellular-connected UAV systems.

To mitigate these challenges, a substantial body of research has been devoted to optimizing UAV flight trajectories to ensure stable cellular connectivity. Early studies primarily adopted simplified statistical channel models to preserve analytical tractability. For instance, [6] and [7] employed graph theory and convex optimization techniques to minimize mission completion time under outage constraints. [8] further explored UAV communications in rural scenarios by transforming outage constraints into spatial flight restrictions and designing a greedy algorithm to minimize flight time. Although these methods demonstrate effectiveness in idealized settings, the reliance on simplified models often fails to capture the complex propagation characteristics encountered in real-world environments.

Recent studies have increasingly adopted advanced channel models and leveraged radio maps to guide UAV trajectory design for stable communication connectivity. For instance, the work in [9] proposed a deep reinforcement learning (DRL) framework for joint navigation and radio mapping, dynamically updating the radio map during flight trajectory. Zhang et al. further incorporated radio maps into a DRL framework to minimize the weighted sum of task completion time and outage duration [10], while Xie et al. employed a Dueling Double Deep Q-Network to optimize energy-efficient 3D trajectories under communication constraints [11]. Similar works have addressed related problems in urban environments [12], [13]. However, most existing studies primarily emphasize mission completion without explicitly considering real-time data transmission requirements. They typically rely solely on ground-to-air (G2A) radio maps to ensure downlink connectivity. In realistic inspection scenarios, UAVs must not only collect but also efficiently upload mission data to the ground network, which necessitates the joint consideration of both G2A maps for downlink control and air-to-ground (A2G) maps for uplink transmission efficiency.

To bridge this gap, this paper presents an innovative framework for cellular-connected UAV inspection systems. The main contributions are summarized as follows,

- We propose a system architecture that jointly incorporates G2A and A2G radio maps to ensure reliable downlink control and efficient uplink data transmission.
- We formulate a comprehensive optimization problem that

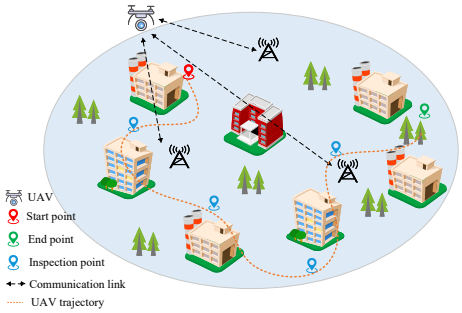


Fig. 1. An illustration of the system architecture.

jointly determines the inspection sequence and flight trajectory, aiming to minimize the total mission completion time and overall transmission delay.

- To solve the formulated optimization problem, we develop a hybrid algorithm that decomposes it into two tractable subproblems. An improved genetic algorithm (IGA) first determines the optimal inspection sequence by jointly considering flight distance, communication throughput, and outage duration. Subsequently, a DRL algorithm refines the UAV's continuous flight trajectory.

II. System Model

Consider a cellular-connected UAV inspection system deployed in a rural area, as illustrated in Fig. 1. A rotary-wing UAV equipped with a high-resolution camera is assigned to visit K inspection points to capture image data, which must be transmitted to GBSs in real time. The UAV departs from an initial location $\mathbf{q}_s$, sequentially visits all inspection points, and ultimately arrives at the destination $\mathbf{q}_e$. To minimize the overall mission duration while maintaining reliable communication, the UAV must jointly optimize the inspection sequence and its flight trajectory, considering the communication coverage and throughput characteristics along potential routes derived from the G2A and A2G radio maps. Let the set of inspection points be denoted as $\mathcal{W} = \{\mathbf{w}_1, \mathbf{w}_2, \cdots, \mathbf{w}_K\}$, where $\mathbf{w}_k = [x_k, y_k]^\top$ represents the horizontal coordinates of the k-th inspection point. The inspection sequence is defined as $\mathcal{O} = \{O_0, O_1, O_2, \ldots, O_K, O_{K+1}\}$, where $O_k \in \{1, 2, \ldots, K\}$ indicates the index of the k-th visited inspection point, with $\mathbf{w}_{O_0} = \mathbf{q}_s$ and $\mathbf{w}_{O_{K+1}} = \mathbf{q}_e$. The UAV is assumed to fly at a fixed altitude H_u, which is higher than the maximum terrain and building heights to prevent potential collisions [5]. Each inspection point requires the collection of an image data packet of size I, and it is assumed to occur instantaneously upon reaching each point.

The total mission duration T is discretized into N time steps with duration Δt. The UAV's horizontal position at time step n is denoted by $\mathbf{q}[n] = [x[n], y[n]]^\top$, and the complete flight trajectory is represented by the sequence $\mathcal{Q} = \{\mathbf{q}[0], \mathbf{q}[1], \ldots, \mathbf{q}[N]\}$, where $\mathbf{q}[0] = \mathbf{q}_s$ and $\mathbf{q}[N] = \mathbf{q}_e$.

A. Communication Model

Unlike existing studies that rely on simplified LoS or probabilistic channel models, this work adopts the more realistic 3GPP Rural Macro (RMa) channel model [14], which accurately characterizes the propagation behavior in rural environments with sparse base stations.

1) Channel Model: The network consists of M cellular sectors (deployed across multiple GBSs). At each time step n, the UAV is allowed to associate with at most one sector by employing time-division multiple access (TDMA) technology.

The channel gain between the UAV and sector m at time step n is modeled as

$$h_m[n] = \sqrt{G_m[n] \cdot \beta_m[n]} \cdot \alpha_m[n] \tag{1}$$

where $G_m[n]$ denotes the antenna gain, $\beta_m[n]$ and $\alpha_m[n]$ represent the large-scale fading coefficient and the small-scale fading coefficient, respectively.

The large-scale fading coefficient $\beta_m[n]$ depends on the LoS condition, given by

$$\beta_m[n] = \begin{cases} \mathrm{PL}_m^{\mathrm{LoS}}[n], & \text{if LoS link,} \\ \mathrm{PL}_m^{\mathrm{NLoS}}[n], & \text{if non-LoS (NLoS) link.} \end{cases} \tag{2}$$

Given the building distribution and terrain characteristics of the area, the LoS condition is determined by detecting the presence of obstacles between the UAV and the serving GBS. According to the 3GPP RMa model [14], the path loss for LoS and NLoS links is given respectively by

$$\begin{aligned} \mathrm{PL}_m^{\mathrm{LoS}}[n] = {} & 20\log_{10}\left(\frac{40\pi d_m[n] f_c}{3}\right) \\ & + \min\left(0.03 H_u^{1.72},\ 10\right)\log_{10}(d_m[n]) \\ & - 0.58\log_{10}(H_g), \end{aligned} \tag{3}$$

$$\begin{aligned} \mathrm{PL}_m^{\mathrm{NLoS}}[n] = {} & 161.04 - 7.1\log_{10}(20) + 7.5\log_{10}(H_g) \\ & - \left(24.37 - 3.7\left(\frac{H_g}{H_u}\right)^2\right)\log_{10}(H_g) \\ & + (43.42 - 3.1\log_{10}(H_g))\,(\log_{10}(d_m[n]) - 3) \\ & + 20\log_{10}(f_c) \\ & - \left[3.2\,(\log_{10}(11.75 H_g))^2 - 4.97\right], \end{aligned} \tag{4}$$

where $d_m[n] = \|\mathbf{q}[n] - \mathbf{g}_m\|$ denotes the distance between the UAV and sector m, H_g represents the GBS height, f_c is the carrier frequency. The small-scale fading coefficient $\alpha_m[n]$ follows a Rician distribution with a Rician factor of 15 dB for LoS links and a Rayleigh distribution for NLoS links [10].

2) G2A Communications: The downlink signal-to-interference-plus-noise ratio (SINR) from sector m to the UAV is given by

$$\gamma_m^{\mathrm{G2A}}[n] = \frac{P_m \cdot |h_m[n]|^2}{\sigma^2 + \sum_{j \neq m} P_j \cdot |h_j[n]|^2}, \tag{5}$$

where P_m is the transmit power of sector m and σ^2 is the noise power. At each time step n, the UAV associates only with the sector m that provides the highest downlink SINR [9]. An outage event occurs when the SINR falls below a predefined threshold $\gamma_{\min}^{\mathrm{G2A}}$. Thus, the outage probability at time step n is expressed as

$$P_{\mathrm{out}}[n] = \Pr\left\{\gamma_m^{\mathrm{G2A}}[n] \leq \gamma_{\min}^{\mathrm{G2A}}\right\}, \tag{6}$$

where $\Pr\{\cdot\}$ denotes the probability of an event, and the total outage duration over the entire mission is given by

$$T_{\text{out}} = \sum_{n=0}^{N} P_{\text{out}}[n] \cdot \Delta t. \tag{7}$$

3) A2G Communications: For uplink transmission, the signal-to-noise ratio (SNR) at the receiving sector m is

$$\gamma_m^{\text{A2G}}[n] = \frac{P_u \cdot |h_m[n]|^2}{\sigma^2}, \tag{8}$$

where P_u is the UAV's transmit power. The corresponding achievable data rate is

$$R_m[n] = B \cdot \log_2\left(1 + \gamma_m^{\text{A2G}}[n]\right), \tag{9}$$

where B denotes the communication bandwidth.

4) Data Offloading Model: Let n_k denote the time step at which the UAV reaches the k-th inspection point O_k. The binary variable $\rho_m[n] \in \{0,1\}$ indicates whether the UAV transmits data to sector m at time step n. During the flight from O_k to O_{k+1}, the UAV offloads the data packet of size I. The amount of data offloaded during this flight segment is

$$D_k = \sum_{n=n_k}^{n_{k+1}-1} \sum_{m=1}^{M} \rho_m[n] \cdot R_m[n] \cdot \Delta t, \tag{10}$$

The transmission delay τ_k for the data collected at point O_k is defined as the minimum time required to transmit I bits, starting from time step n_k, which is calculated by

$$\tau_k = \min\left\{ \tau \in \mathbb{N} \;\middle|\; \sum_{m=1}^{M} \rho_m[n] R_m[n] \cdot \tau \cdot \Delta t \geq I \right\}. \tag{11}$$

The total transmission delay is then given by

$$\tau_{\text{total}} = \sum_{k=1}^{K} \tau_k \cdot \Delta t. \tag{12}$$

B. Problem Formulation

We formulate a joint optimization problem over the UAV trajectory $\mathcal{Q}$, transmission scheduling $\boldsymbol{\rho} = \{\rho_m[n]\}$, and inspection sequence $\mathcal{O}$, aiming to minimize the sum of mission completion time and overall transmission delay, given by

$$\text{(P0):} \quad \min_{\mathcal{Q},\mathcal{O},\boldsymbol{\rho}} \quad T + \tau_{\text{total}} \tag{13}$$

$$\text{s.t.} \quad \begin{cases} \sum_{m=1}^{M} \rho_m[n] \leq 1,\ \rho_m[n] \in \{0,1\},\ \forall m,n & \text{(14a)} \\ D_k \geq I, \quad \forall k = 1,\ldots,K & \text{(14b)} \\ T_{\text{out}} \leq T_{\text{out}}^{\max} & \text{(14c)} \\ \|\mathbf{q}[n+1] - \mathbf{q}[n]\| \leq v_{\max}\Delta t,\ \forall n & \text{(14d)} \\ \mathbf{q}[0] = \mathbf{q}_s, \quad \mathbf{q}[N] = \mathbf{q}_e & \text{(14e)} \\ 0 \leq x[n] \leq D,\ 0 \leq y[n] \leq D,\ \forall n & \text{(14f)} \end{cases}$$

The constraint (14a) ensures that the UAV associates with only one sector at each time step. Constraint (14b) guarantees the complete offloading between consecutive inspection points, while constraint (14c) restricts the maximum allowable outage duration. Constraint (14d) enforces the maximum speed limit, and constraints (14e) and (14f) specify the boundary conditions and operational area, respectively.

Problem (P0) is challenging to solve directly due to the strong coupling among continuous trajectory variables, discrete scheduling decisions, and combinatorial inspection sequence optimization, as well as the non-convex nature of constraints (14b) and (14c). Furthermore, the inspection sequence optimization itself is NP-hard, complicating the problem.

III. DRL-IGA DESIGN

This section presents the proposed DRL-IGA, i.e., the DRL algorithm enhanced by IGA, to address the joint optimization problem (P0). The proposed framework decomposes the original problem into two tractable subproblems. First, the inspection sequence optimization is modeled as a traveling salesman problem (TSP) and solved using a novel IGA. Subsequently, conditioned on the optimal sequence, the UAV's continuous flight trajectory is refined through the twin delayed deep deterministic policy gradient (TD3) algorithm.

A. Inspection Sequence Optimization via IGA

The inspection sequence has a substantial impact on both the total flight distance and the communication quality experienced throughout the mission. A well-planned sequence can markedly reduce mission completion time and mitigate transmission delays. Although the problem of determining the visiting order of inspection points can be abstracted as a TSP, conventional TSP formulations primarily aim to minimize total travel distance, neglecting communication-related factors. This simplification is inadequate, as communication quality is equally critical for ensuring reliable data transmission.

To overcome this limitation, we design an IGA that jointly considers flight distance, uplink communication throughput, and downlink outage duration. We begin by constructing an undirected graph $\mathcal{G} \triangleq (V, E)$ representing the spatial topology of the inspection points. The vertex set V is defined as

$$V = \{\mathbf{q}_s, \mathrm{w}_1, \mathrm{w}_2, \cdots, \mathrm{w}_K\}. \tag{15}$$

The edge set E is represented by

$$E = \{\psi_{ij}, i \neq j, i, j \in V\}. \tag{16}$$

The edge weight ψ_{ij} is designed to capture the trade-off between flight efficiency and communication quality. Specifically, to minimize downlink outage duration, the UAV should avoid regions with weak G2A coverage, whereas to improve uplink transmission efficiency, it should prioritize areas with high A2G SNR to facilitate faster data offloading. Accordingly, a composite metric that jointly considers the flight distance $d_{i,j}$, the estimated uplink throughput $C_{i,j}$, and the estimated downlink outage time $t_{i,j}$ is defined by ψ_{ij}, which is given by

$$\psi_{ij} = \begin{cases} \frac{C_{i,j}}{d_{i,j} \cdot t_{i,j}}, & \forall i,j \in V,\ i \neq j \\ \infty, & i = j, \end{cases} \tag{17}$$

Algorithm 1 DRL-IGA for UAV Trajectory and Inspection Sequence Optimization

Initialize: Inspection points $\mathcal{W}$, start point $\mathbf{q}_s$, end point $\mathbf{q}_e$, radio maps (G2A, A2G), and data size I
for each generation **do**
    Evaluate fitness of each sequence
    Select top individuals for crossover and mutation
    Generate new population
end for
Obtain optimized inspection sequence $\mathcal{O}^*$
for each episode **do**
    Reset UAV state $\mathbf{q}_0$
    for each time step $t = 0, 1, \ldots, T$ **do**
        Select action $a_t = \mu(s_t) + \mathcal{N}_t$
        Execute a_t, update state, and compute reward r_t
        Store transition (s_t, a_t, r_t, s_{t+1}) in $\mathcal{B}$
        Sample in $\mathcal{B}$, update critic and actor networks
    end for
end for
Derive optimized trajectory $\mathcal{Q}^*$ from trained policy
return $\mathcal{O}^*$, $\mathcal{Q}^*$

where $d_{i,j}$ denotes the Euclidean distance between inspection points i and j, $C_{i,j}$ represents the estimated uplink communication throughput along the flight path from i to j, $t_{i,j}$ is the estimated downlink outage duration during that flight segment.

Based on the defined edge weight, the inspection sequence optimization problem can be formulated as a variant of TSP, given by

$$\text{(P1):} \quad \min \sum_{i=0}^{K} \sum_{j=0}^{K} \psi_{ij} \Phi_{ij}, \tag{18}$$

where Φ_{ij} is a binary variable that $\Phi_{ij} = 1$ if the UAV travels directly from point i to point j, and $\Phi_{ij} = 0$ otherwise. The standard genetic algorithm is then applied to solve problem (P1) and determine the optimal inspection sequence $\mathcal{O}$.

B. Optimal Flight Trajectory based on DRL

With the UAV trajectory discretized into N time steps, the position at each step defines the system state. The UAV's kinematic model in the Cartesian coordinate system is described as

$$\begin{aligned} x[n+1] &= x[n] + v[n]\Delta t \cos(\theta[n]), \\ y[n+1] &= y[n] + v[n]\Delta t \sin(\theta[n]), \end{aligned} \tag{19}$$

where $v[n] \in [0, v_{\max}]$ and $\theta[n] \in [0, 2\pi]$ represent the speed and heading angle at time step n, respectively.

Given the optimal inspection sequence $\mathcal{O}$ obtained previously, we now focus on optimizing the UAV flight trajectory $\mathcal{Q}$ and transmission scheduling $\rho_m[n]$. To ensure that data transmission is completed before reaching the next inspection point, the UAV adopts a best-channel association strategy, whereby it transmits to the sector offering the highest uplink SNR $\gamma_m^{\text{A2G}}[n]$ to maximize the achievable data rate. This scheduling policy is formalized as

$$\rho_m[n] = \begin{cases} 1, & \text{if } m = \arg\max\limits_{m} \gamma_m^{\text{A2G}}[n], \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

With the transmission scheduling determined, the original problem (P0) is reduced to optimizing only the UAV trajectory $\mathcal{Q}$. To address this continuous control problem using deep reinforcement learning, it is reformulated as a Markov Decision Process (MDP), denoted as

$$\begin{aligned} \text{(P2):} \quad & \min_{\mathcal{Q}} \; T + \tau_{\text{total}} \\ & \text{s.t.} \quad \text{(14a)} - \text{(14f)}. \end{aligned} \tag{21}$$

The MDP is characterized by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$, where each component is defined as

- *State* $\mathcal{S}$: The state at time step n is defined as $s_n = [\mathbf{q}_n, \mathbf{I}_n, \Gamma_n]$, where $\mathbf{q}_n$ denotes the UAV's current location, $\mathbf{I}_n$ represents the amount of data currently stored on the UAV, and Γ_n specifies the coordinates of the target inspection point.
- *Action* $\mathcal{A}$: The action at time step n is defined as $a_n = [v_n, \theta_n]$, where $v_n \in [0, v_{\max}]$ is the UAV's velocity and $\theta_n \in [0, 2\pi]$ denotes its heading angle.
- *Reward* $\mathcal{R}$: The reward quantifies the quality of the agent's action in a given state. To transform the multi-objective optimization problem into a reinforcement learning framework, the reward function is designed to maximize the cumulative return. Since the objective of this study is to minimize the sum of mission completion time and total transmission delay, the reward function is defined as

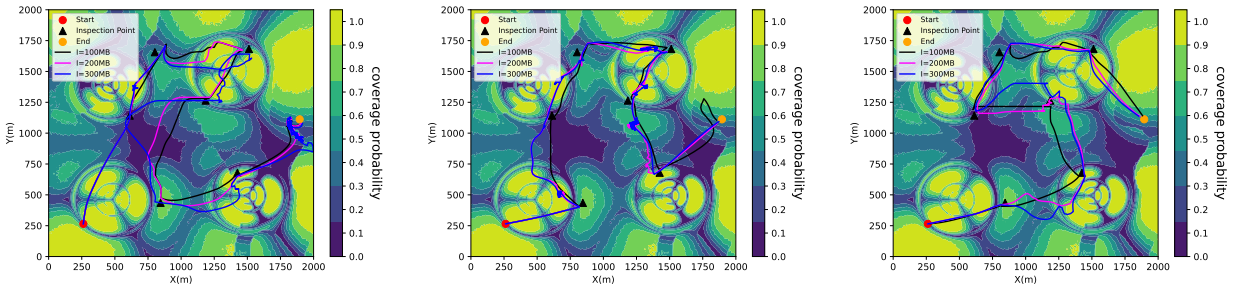
$$r_n = R_{arrive} + R_{done} - P_{trans} - P_{out}, \tag{22}$$

  where R_{arrive} is the reward for reaching the target inspection point, R_{done} represents the reward for completing the inspection task, P_{trans} is the penalty imposed for failing to complete data transmission before reaching the target point, and P_{out} is the penalty for flying through areas of low coverage probability.

Subsequently, the problem (P0) can be addressed using a DRL method. In this work, the TD3 algorithm is adopted, as it effectively handles continuous state and action spaces while enhancing training stability and performance, making it particularly suitable for UAV trajectory optimization tasks [11]. The detailed procedure of the proposed DRL-IGA is summarized in Algorithm 1.

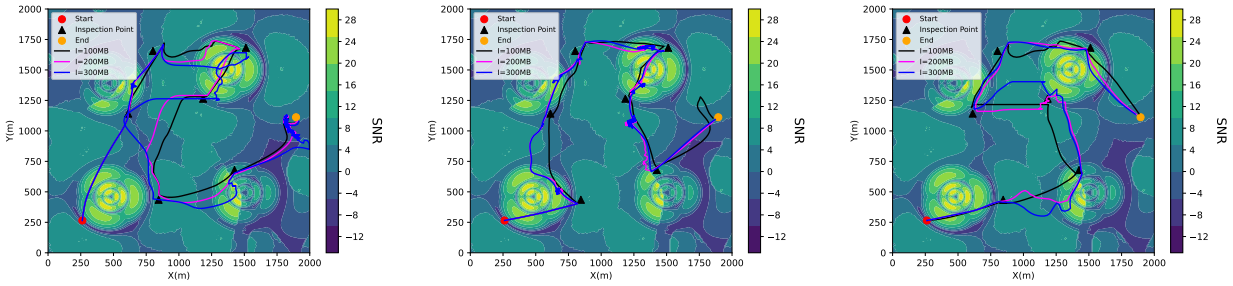
IV. Simulation Results

In this section, we present simulation results to evaluate the performance of the proposed DRL-IGA. As illustrated in Fig. 1, a cellular-connected UAV inspection system is considered, where 4 GBSs and $K = 6$ inspection points (excluding the start and end point) are deployed within a rural area of size $2 \text{ km} \times 2 \text{ km}$. Each GBS is equipped with three sectors, resulting in a total of $M = 12$ sectors. The carrier frequency is



(a) TSP (b) DRL-SA (c) Proposed Algorithm

Fig. 2. Trajectory comparison in the G2A radio map.



(a) TSP (b) DRL-SA (c) Proposed Algorithm

Fig. 3. Trajectory comparison in the A2G radio map.

$f_c = 2$ MHz, the system bandwidth is $B = 1$ MHz, and noise power is $\sigma^2 = -100$ dBm. The transmit powers of the GBSs and UAV are $P_m = 0.15$ W and $P_u = 0.1$ W, respectively. The maximum tolerated outage duration is $T_{\text{out}}^{\max} = 80$ s, and the UAV's maximum flight speed is $v_{\max} = 20$ m/s. The UAV flight altitude and GBS antenna height are set to $H_u = 100$ m and $H_g = 25$ m, respectively. The building distribution follows the statistical model in [9] and [10], where the ratio of building area to total area is $\alpha_{bd} = 0.2$, the mean number of buildings per unit area is $\beta_{bd} = 100$, the building height distribution $\sigma_{bd} = 30$, and the maximum building height is 50 m. To demonstrate the effectiveness of the proposed method, two benchmark schemes are selected for comparison, which are described as follows

- TSP: In this scheme, the inspection sequence is obtained by solving the TSP that considers only the flight distance, without leveraging G2A and A2G radio maps. Subsequently, a standard DRL method is employed to optimize the UAV's flight trajectory.
- DRL with simulated annealing algorithm (DRL-SA) [2]: In this scheme, the simulated annealing algorithm is used to determine the inspection sequence by jointly considering both flight distance and downlink outage duration based on the G2A radio map. Then the TD3 algorithm is used for UAV trajectory optimization.

A. Trajectory Analysis under Varying Data Sizes

The optimized UAV trajectories under varying data sizes I and algorithmic schemes are illustrated in Fig. 2 and Fig. 3, superimposed upon G2A and A2G radio maps depicting downlink coverage probability and uplink SNR distribution, respectively.

Figure 2 reveals a significant advantage of utilizing the G2A radio map. The trajectories generated by the TSP-based scheme frequently traverse regions exhibiting poor or nearly zero downlink coverage probability, as it does not utilize the G2A radio map to estimate the outage duration during inspection sequence planning. In contrast, the trajectories generated by the DRL-SA algorithm effectively avoid most low-coverage regions, although they still pass through a few such areas in the later stages of the mission. The proposed DRL-IGA, however, generates trajectories that almost entirely bypass low-coverage zones. This improvement arises from the joint consideration of outage duration in the edge weight computation using the G2A radio map, which proactively guides the UAV toward regions with stronger downlink and enhances overall communication robustness.

Figure 3 illustrates a notable drawback of not utilizing the A2G radio map. When $I = 100$ MB, the trajectories of all three methods remain relatively direct, as the transmission tasks can be completed rapidly even under moderate data rates. However, as the data load increases to $I = 200$ MB and $I = 300$ MB, the trajectories generated by the TSP and DRL-SA schemes become discernibly more circuitous, with the UAV deviating toward regions of higher SNR, sometimes even hovering in these areas to improve uplink throughput and mitigate transmission latency. In contrast, the trajectories generated by the proposed DRL-IGA exhibit only minor deviations despite moderate route adjustments. This efficiency stems from the algorithm's use of the A2G radio map to estimate uplink data rates during inspection sequence planning, enabling the UAV to consistently follow paths with higher data rates and reduced transmission delay.

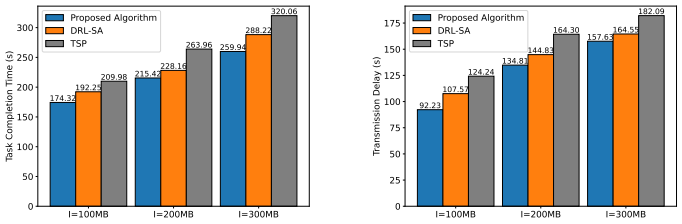


Fig. 4. Comparison of task completion time and transmission delay.

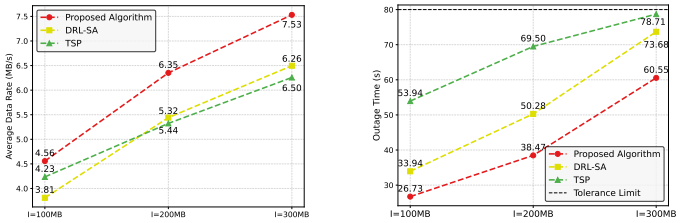


Fig. 5. Comparison of uplink average data rate and downlink outage duration.

B. Quantitative Performance Comparison

Quantitative evaluation in Fig. 4 substantiates the performance superiority of the proposed algorithm in terms of total mission completion time and transmission delay. Both metrics exhibit a monotonic increase with the data size I across all methods. Notably, the proposed DRL-IGA maintains a consistent and significant advantage, yielding the lowest values of both T and τ_{total} under all considered conditions. This performance gap is quantitatively highlighted at $I = 300$ MB, where the proposed DRL-IGA achieves a reduction in the composite objective $T + \tau_{\text{total}}$ of approximately 10% compared with DRL-SA and exceeds 20% against the TSP scheme.

A comparative analysis of the uplink average data rates and downlink outage duration is provided in Fig. 5. Specifically, the extended outage duration of the TSP scheme is attributable to its neglect of the G2A radio map. While the DRL-SA scheme successfully leverages the G2A radio map to mitigate outages, it fails to incorporate the A2G radio map, which is critical for uplink channels scheduling. This fundamental limitation in both baseline schemes results in their substantially lower uplink data rates. In contrast, the proposed DRL-IGA demonstrates a balanced optimization by jointly exploiting the A2G and G2A radio maps. This dual-map exploitation strategy enables the concurrent maximization of the uplink data rates and the minimization of the downlink outage time.

V. Conclusion

This paper investigated the joint optimization of flight trajectory and inspection sequence for a cellular-connected UAV inspection system in rural environments. The objective was to minimize the overall mission completion time and data transmission delay while maintaining reliable communication connectivity. To address the resulting mixed-integer and non-convex optimization problem, we proposed a novel algorithm called DRL-IGA to decompose the original problem into two tractable subproblems. First, IGA determines an optimal inspection sequence by jointly considering flight distance, uplink communication throughput, and downlink outage duration. Subsequently, the TD3 algorithm optimizes the UAV's continuous flight trajectory based on the obtained sequence. Simulation results show that the proposed DRL-IGA significantly outperforms baseline schemes in terms of mission efficiency, transmission delay, and communication reliability, demonstrating its effectiveness for intelligent and resilient UAV-assisted inspection systems. Future work will extend the framework to multi-UAV collaboration, real-time radio map updates, and energy-efficient task allocation for improved overall performance in dynamic environments.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under Grant Nos. 62301411, 62171352, U22A2029, and U24A20237.

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